

ARUN CHANDANA

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Professional Summary

Manufacturing Engineer with experience in process optimization, Lean Manufacturing, and quality control within regulated environments. Seeking to leverage knowledge of manufacturing processes, data analysis, and continuous improvement methodologies to enhance efficiency and quality in a dynamic manufacturing setting. Familiar with CGMP principles and applications within the manufacturing process. Experienced with optimizing manufacturing processes to improve productivity and reduce waste. Utilizes technical expertise to streamline operations and ensure seamless workflow, knowledge of quality control and process automation, contributing to continuous improvement and operational excellence.

Experience

BABA Technocrats and Manufacturers Pvt Ltd

Nov 2021 - Mar 2023

MANUFACTURING ENGINEER AND QUALITY ENGINEER

Vijayawada, India

- Optimized production planning and scheduling to align workflows with operational goals, enhancing efficiency and reducing downtime
- Developed and maintained Standard Operating Procedures (SOPs) to ensure consistent product quality and smooth integration of new processes
- Implemented Statistical Process Control (SPC) to monitor process variations, improving quality control by 18%
- Engaged in Kaizen events to identify inefficiencies and drive continuous improvement, achieving a 25% reduction in waste.

Lakshmi Chaitanya Alloys

Nov 2018 - Apr 2019

MANUFACTURING ENGINEER AND QUALITY ENGINEER

Hyderabad, India

- Conducted quality inspections and executed corrective actions to reduce defect rates by 20% while maintaining compliance with quality systems
- Optimized casting processes through the application of lean manufacturing principles and rigorous process optimization techniques
- Performed root cause analysis (RCA) on product defects, reducing non-conformities and rework by 15%
- Ensured adherence to ISO standards via regular process audits and comprehensive documentation.
- Provided technical support to troubleshoot equipment issues, minimizing downtime and maintaining production targets.
- Improved production efficiency by developing and implementing new manufacturing processes.
- Increased production volume, optimizing equipment layouts for maximum efficiency.

Core Qualifications

- Manufacturing Processes:** Lean Manufacturing High Mix/Volume Manufacturing, Equipment Qualification & Routings, Process Optimization, Manufacturing Processes, Process Design & Optimization, Industrial & Mechanical Engineering Principles
- Process Improvement:** Six Sigma(Green Belt), Kaizen, Audit Process Evaluation, Kaizen events, 5S, Efficiency Improvement, process management systems, Efficiency Improvements, Labor Standards & Work Measurement, Cost Reduction, labor standards, packaging, fixtures
- Quality & Standards:** ISO 9001 Standards, Standard Operating Procedures, Quality Control Proficiency, Standard Work, Control Plans and Statistical Process Control, Design of Experiments (DOE), Root-Cause Analysis (FMEA, DMAIC), Audit Support & Continuous Monitoring, PLC, controls engineering, Data Analysis, Quality assurance
- Technical & Analytical Skills::** Task Management Instructions, SQL (Basic), R, Data Visualization (Matplotlib, seaborn), Python (NumPy, pandas, scikit-learn), IoT devices, robots, software tools, IoT Device Integration & Digital-Twin Simulation, programming knowledge, PLC/Controls Engineering & Robot Programming, CNC Programming
- Project & People Management::** Cross-Functional Team Leadership, Training & Standard Work Development, Project Planning, Budgeting & Execution, Real-Time Performance Monitoring (OEE, KPIs)

Education

Texas Tech University, Quality Control - Lubbock, Texas

May 2025

Master of Engineering, Manufacturing Systems, Industrial Engineering

- GPA: 3.333

Seshadri Rao Gudlavalleru Engineering College - Gudlavalleru, India

Mar 2022

Bachelor of Science, Mechanical Engineering

A.A.N.M. & V.V.R.S.R. POLYTECHNIC - Gudlavalleru, India

Jun 2019

Diploma, Mechanical Engineering

Projects

IoT-Integrated Drone System, IoT Developer

Texas Tech University

Lubbock, TX

- Designed an IoT-integrated drone system featuring 5G/6G, Wi-Fi, and LoRa WAN connectivity with onboard edge processing; developed seamless cloud integration for real-time data analytics; authored technical and economic feasibility reports projecting a 57.5% CAGR market opportunity and a 30% gross margin on hardware.

Chick-fil-A Student Union Queueing Simulation

Texas Tech University

Lubbock, TX

- Collected and fitted arrival/service data in Arena, modeled multi-server scenarios under 25%/50% demand surges, and recommended a three-server configuration that cut wait times by 30%.

Automated Pick-and-Place Cell with Dobot MG400

Texas Tech University

Lubbock, TX

- Programmed the Dobot MG400 and Siemens S7-1200 PLC, developed error-proofing ladder logic, and validated the cell in Simio, achieving a 25% throughput increase and sub-0.5 mm placement accuracy.

Fracture Mechanics Analysis of Metallic Components

Texas Tech University

Lubbock, TX

- Performed ANSYS FEA to compute stress-intensity factors, applied Paris' law for fatigue-life prediction, and demonstrated a 20% safety margin over baseline operating loads.